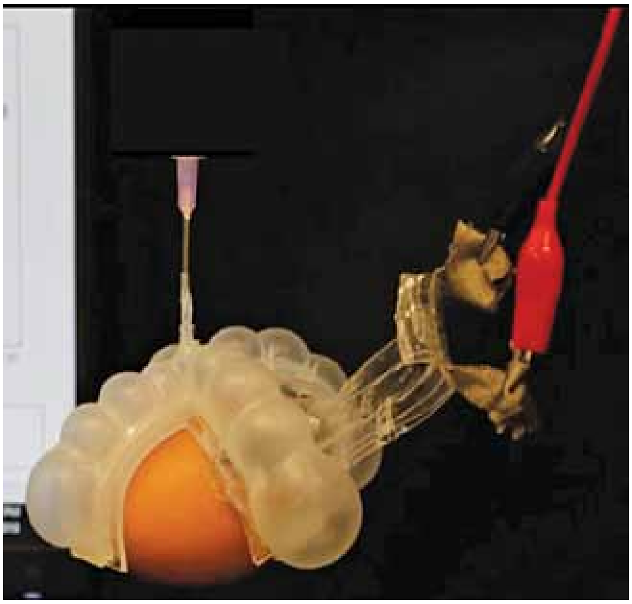
Soft, stretchable, self-powered thermometer

Researchers at the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS) have developed a soft, stretchable, self-powered thermometer that can be integrated

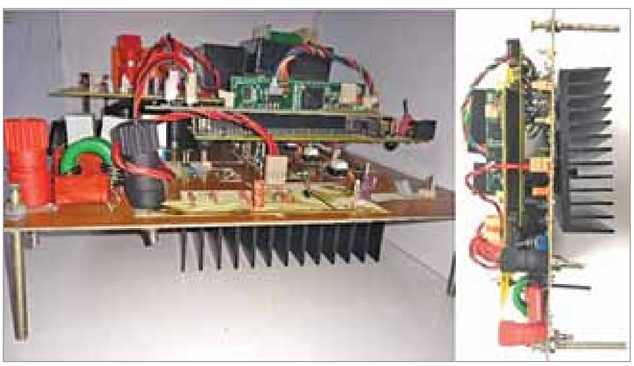


A stretchable thermometer for soft robots (Credit: www.seas.harvard.edu)

into stretchable electronics and soft robots. The thermometer consists of three simple parts: an electrolyte, an electrode, and a dielectric material to separate the two. The electrolyte/dielectric interface accumulates ions while the dielectric/electrode interface accumulates electrons. The charge imbalance between the two sets up an ionic cloud in the electrolyte. When the temperature changes, the ionic cloud changes thickness, and a voltage is generated. The voltage is sensitive to temperature but insensitive to stretch.

On-board chargers for EVs

A team of researchers at IIT-BHU Varanasi, along with experts from IIT Guwahati and IIT Bhubaneswar, have devel-



EV charging prototype (Credit: www.electronicsforu.com)

oped a new, low-cost technology for the on-board charging of vehicles that can further bring down costs of two- and four-wheeler electric vehicles. The technology will also enhance the charging infrastructure of vehicles in India and contribute towards sustainable development of EVs by improving community health through the elimination of tailpipe emissions and reduced reliance on fossil fuels.

Face mask turned into smart monitoring device

Northwestern University engineers have developed a new smart sensor platform for face masks that they are calling a ‘Fitbit for the face.’ Dubbed ‘FaceBit,’ the lightweight,

Inside of a mask with FaceBit device (Credit: https://news.northwestern.edu)

quarter-sized sensor uses a tiny magnet to attach to any N95, cloth, or surgical face mask. Not only can it sense the user’s real-time respiration rate, heart rate, and mask wear time, it also may be able to replace cumbersome tests. All the information is wirelessly transmitted to a smartphone app, which contains a dashboard for real-time health monitoring. The app can immediately alert the user when issues such as elevated heart rate or a leak in the mask unexpectedly arise. The physiological data could also be used to predict fatigue, physical health status, and emotional state.

Glucose sensor based on boronic acids and graphene

Researchers at the University of Bath in collaboration with industrial partner, Integrated Graphene, have developed a new sensing technique based on graphene foam for the detection of glucose levels in the blood. Since it is a chemical